Case Study 2: Employee Payroll and Deduction Analysis

1. Problem Statement

To develop a Python program to calculate an employee's gross salary based on basic pay, allowances, overtime, and incentives. The program applies deductions and generates a formatted salary statement showing gross salary, total deductions, and net salary.

2. Objectives

- To calculate the gross salary of an employee.
- To calculate allowances, overtime, and incentives.
- To apply salary deductions.
- To calculate the final net salary.
- To display a formatted salary statement.

3. Assumptions

For this assignment, we use the following sample values. They are assumed values for learning purposes, not official salary or tax rules.

Component	Amount or rule
Basic salary	Entered by the user
House Rent Allowance (HRA)	20% of basic salary
Other allowance	10% of basic salary

Overtime	₹100 per hour
Incentive	Entered by the user
Deduction	5% of gross salary
Professional deduction	₹200 if gross salary is above ₹20,000

Formula:

Gross Salary = Basic Salary + HRA + Other Allowance + Overtime Pay + Incentive

Net Salary = Gross Salary − Total Deductions

4. Algorithm

1. Start the program.
2. Enter the employee's name and basic salary.
3. Enter overtime hours and incentive amount.
4. Calculate HRA and other allowances.
5. Calculate overtime pay.
6. Calculate gross salary.
7. Calculate deductions based on the given conditions.
8. Calculate net salary.
9. Display the complete salary statement.
10. Stop the program.

5. Flowchart

Start
Input employee details, basic salary, overtime hours and incentive
Calculate HRA, other allowance and overtime pay
Calculate gross salary
Calculate deductions and net salary
Display salary statement
Stop

6. Pseudocode

```
None
START

INPUT employee_name
INPUT basic_salary
INPUT overtime_hours
INPUT incentive

HRA = basic_salary * 20 / 100
other_allowance = basic_salary * 10 / 100
overtime_pay = overtime_hours * 100

gross_salary = basic_salary + HRA
               + other_allowance
               + overtime_pay + incentive

deduction = gross_salary * 5 / 100

IF gross_salary > 20000:
    deduction = deduction + 200

net_salary = gross_salary - deduction

DISPLAY employee salary statement

STOP
```

7. Complete Python source code

```
None

# Employee Payroll and Deduction Analysis
```

```
employee_name = input("Enter employee name: ")
basic_salary = float(input("Enter basic salary: "))
overtime_hours = float(input("Enter overtime hours: "))
incentive = float(input("Enter incentive amount: "))

# Validate input
if basic_salary < 0 or overtime_hours < 0 or incentive < 0:
    print("Invalid input! Values cannot be negative.")

else:
    # Calculate allowances
    hra = basic_salary * 20 / 100
    other_allowance = basic_salary * 10 / 100

    # Calculate overtime pay
    overtime_pay = overtime_hours * 100

    # Calculate gross salary
    gross_salary = (
        basic_salary
        + hra
        + other_allowance
        + overtime_pay
        + incentive
    )

    # Calculate deductions
    deduction = gross_salary * 5 / 100

    if gross_salary > 20000:
        deduction = deduction + 200

    # Calculate net salary
    net_salary = gross_salary - deduction

    # Display salary statement
```

```
print("\n========== SALARY STATEMENT ==========")
print("Employee Name       :", employee_name)
print("Basic Salary        : Rs.", round(basic_salary, 2))
print("HRA                 : Rs.", round(hra, 2))
print("Other Allowance     : Rs.", round(other_allowance, 2))
print("Overtime Pay        : Rs.", round(overtime_pay, 2))
print("Incentive           : Rs.", round(incentive, 2))
print("--------------------------------------")
print("Gross Salary        : Rs.", round(gross_salary, 2))
print("Total Deductions    : Rs.", round(deduction, 2))
print("Net Salary          : Rs.", round(net_salary, 2))
print("======================================")
```

8. Test cases and expected output

Use these sample inputs:

Input	Value
Employee name	Ravi
Basic salary	₹20,000
Overtime hours	5
Incentive	₹1,000

Calculation:

Salary component	Amount
Basic salary	₹20,000
HRA (20%)	₹4,000
Other allowance (10%)	₹2,000
Overtime (5 × ₹100)	₹500
Incentive	₹1,000
Gross salary	₹27,500
5% deduction	₹1,375
Additional deduction	₹200
Net salary	₹25,925

Expected output:

```
None
========== SALARY STATEMENT ==========
Employee Name       : Ravi
Basic Salary        : Rs. 20000.0
```

```
HRA                 : Rs. 4000.0
Other Allowance     : Rs. 6000.0
Overtime Pay        : Rs. 500.0
Incentive           : Rs. 1000.0
--------------------------------------
Gross Salary        : Rs. 33500.0
Total Deductions    : Rs. 1875.0
Net Salary          : Rs. 31625.0
======================================
```

9. Output screenshot

```
======= RESTART: C:/Users/YOGITHA/OneDrive/
Enter employee name: Ravi
Enter basic salary: 20000
Enter overtime hours: 5
Enter incentive amount: 1000

========== SALARY STATEMENT ==========
Employee Name       : Ravi
Basic Salary        : Rs. 20000.0
HRA                 : Rs. 4000.0
Other Allowance     : Rs. 2000.0
Overtime Pay        : Rs. 500.0
Incentive           : Rs. 1000.0
--------------------------------------
Gross Salary        : Rs. 27500.0
Total Deductions    : Rs. 1575.0
Net Salary          : Rs. 25925.0
======================================
```

```python
# Employee Payroll and Deduction Analysis

employee_name = input("Enter employee name: ")
basic_salary = float(input("Enter basic salary: "))
overtime_hours = float(input("Enter overtime hours: "))
incentive = float(input("Enter incentive amount: "))

# Validate input
if basic_salary < 0 or overtime_hours < 0 or incentive < 0:
    print("Invalid input! Values cannot be negative.")

else:
    # Calculate allowances
    hra = basic_salary * 20 / 100
    other_allowance = basic_salary * 10 / 100

    # Calculate overtime pay
    overtime_pay = overtime_hours * 100

    # Calculate gross salary
    gross_salary = (
        basic_salary
        + hra
        + other_allowance
        + overtime_pay
        + incentive
    )

    # Calculate deductions
    deduction = gross_salary * 5 / 100

    if gross_salary > 20000:
        deduction = deduction + 200

    # Calculate net salary
    net_salary = gross_salary - deduction

    # Display salary statement
    print("\n========== SALARY STATEMENT ==========")
    print("Employee Name       :", employee_name)
    print("Basic Salary        : Rs.", round(basic_salary, 2))
    print("HRA                 : Rs.", round(hra, 2))
    print("Other Allowance     : Rs.", round(other_allowance, 2))
    print("Overtime Pay        : Rs.", round(overtime_pay, 2))
    print("Incentive           : Rs.", round(incentive, 2))
    print("--------------------------------------")
    print("Gross Salary        : Rs.", round(gross_salary, 2))
    print("Total Deductions    : Rs.", round(deduction, 2))
    print("Net Salary          : Rs.", round(net_salary, 2))
    print("======================================")
```



10. Conclusion

The Employee Payroll and Deduction Analysis program calculates gross salary, allowances, overtime pay, incentives, and deductions. It determines the net salary and displays a formatted salary statement. This program demonstrates Python variables, arithmetic operators, conditional statements, input, and output